



EOSDIS

NASA'S EARTH OBSERVING SYSTEM
DATA AND INFORMATION SYSTEM

Summary: Cloud-optimized access to HDF5/NetCDF4 files using the Zarr API and the OPeNDAP DMR++ Earthdata Cloud Services

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Cloud-Optimized =

Able to fetch lots of data in a small number of reads

But operating at a scale where you can't just fetch everything

Conclusions

Cloud access optimization demands read latency optimization

1. Combine many smaller reads into fewer larger reads (metadata & data)
2. Skip expensive network actions such as auth or redirects, if possible, via cookies and caching
3. Parallelize reads

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